Hall Ticket Number:

MR-22

Code: B324CS5S

MAHATMA GANDHI INSTITUTE OF TECHNOLOGY (Autonomous)

B.Tech. III Semester Supplementary Examinations, July/August-2024

Computer Organization and Architecture

(Common to CSE, CSB, CSM and CSD)

Time: 3 Hours | Max. Marks: 60

Answer ALL Questions

Part-A (10 x 1 = 10 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
1. a)	Explain Von-Neumann Architecture with neat diagram.	1	2	1	1
1. b)	Distinguish Instruction Cycle and Machine Cycle.	1	4	1	1
1. c)	Explain in-detail about all data transfer and manipulation instructions.	1	2	2	1
1. d)	Explain basic organization of control unit with diagram.	1	2	2	2
1. e)	Explain 2's complement.	1	2	3	2
1. f)	Explain Decimal Arithmetic unit.	1	2	3	3
1. g)	Explain types of memories.	1	2	4	1
1. h)	Differentiate Synchronous and Asynchronous data transmission (any 2 differences).	1	4	4	2
1. i)	Explain Pipelining.	1	2	5	2
1. j)	Explain parallel processing.	1	2	5	1

Part-B (5 x 10 = 50 Marks)

Q. No.	Stem of the Question	M	L	CO	PO
2. a)	Explain the block diagram of digital computer 4bit arithmetic circuit?	5	3	1	3
2. b)	Describe the various registers in a typical CPU with a neat diagram.	5	3	1	2
	OR				
2. c)	Explain the following. i) Conditional branching micro program. ii) Micro instruction instruction format.	5	2	1	1
2. d)	Draw and explain single bus structure.	5	2	1	2
3. a)	Explain Addressing modes in-detail with examples	5	2	2	2
3. b)	List the types of Arithmetic instructions and explain it.	5	2	2	3
	OR				
3. c)	Explain the concept of micro programmed control unit.	5	2	2	1
3. d)	Explain the design of Control unit with a neat diagram.	5	2	2	2
4. a)	Perform the arithmetic operations (+42) +(-13) and (-42) -(-13) in binary using signed 2's complement representation for negative numbers.	5	3	3	4
4. b)	Explain the difference between micro programmed control and hardwired control.	5	3	3	1
	OR				
4. c)	Explain the fixed and floating type numbers that are represented in computers with example.	5	2	3	1
4. d)	Suppose two numbers located in memory are to be added. What are the functional units of digital computer system will carry out this? Explain.	5	2	3	2
5. a)	Explain input/output operations of computer architecture.	5	2	4	1
5. b)	Discuss the different mapping techniques used in cache memories and their relative merits and demerits.	5	2	4	2

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	OR				
5. c)	How can you enhance the speed and capacity of memories? Explain.	5	2	4	2
5. d)	Explain the Direct Memory Access. How it improves the performance of the system?	5	2	4	3
6. a)	Differentiate CISC and RISC Architectures.	5	4	5	1
6. b)	Explain Arithmetic pipeline with suitable diagram.	5	2	5	2
	OR				
6. c)	Explain the characteristics of Multiprocessor?	5	2	5	2
6. d)	Explain in-detail about vector processing and array processing.	5	2	5	3

M: Marks; L: Bloom's Taxonomy Level; CO: Course Outcome; PO: Programme Outcome